

The empirical recurrence for OEIS A230691: recovering a conjecture that is not written down

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7 September 2026

Abstract

OEIS A230691 records “Empirical recurrence of order 94”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 152$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 94, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 94 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A230691 is “Number of $(n+2) \times (2+2)$ 0..2 arrays $x(i,j)$ with each element horizontally, vertically, diagonally or antidiagonally next to at least two elements with value $(x(i,j)+1) \bmod 3$, and upper left element zero.”. It has offset 1 and begins

4, 12, 144, 2494, 39504, 697059, 12059186, 208611750, ...

The entry states:

Empirical recurrence of order 94 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 05:57:03, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4768$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 152$ of the 4768, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 152$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 9536$ exact terms of the model returns order 94 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{94} c_i a(n-i),$$

c_1	20	c_{25}	−10836545804417	c_{49}	−179455553638961544	c_{73}	−27527407263320291
c_2	−44	c_{26}	7031245662739	c_{50}	−125805296464715012	c_{74}	20138595699993358
c_3	225	c_{27}	3687611955164	c_{51}	209927283350646454	c_{75}	−7202753787989221
c_4	−5511	c_{28}	36679005742778	c_{52}	8134317333855907	c_{76}	1492351924142205
c_5	16420	c_{29}	−60304353664802	c_{53}	7769368577533176	c_{77}	1141839644688821
c_6	−28491	c_{30}	−128479740113269	c_{54}	110219092530578465	c_{78}	302744578886859
c_7	505116	c_{31}	63546428101775	c_{55}	−331133714107024378	c_{79}	−230861678797648
c_8	−1470168	c_{32}	248120131111721	c_{56}	108670858494273031	c_{80}	−26332456442506
c_9	819983	c_{33}	665867062741594	c_{57}	510420897671952024	c_{81}	−10813417562611
c_{10}	−20013601	c_{34}	−861124190945694	c_{58}	−647437605765282934	c_{82}	16482352426647
c_{11}	55997184	c_{35}	−820162242340315	c_{59}	531733270785453748	c_{83}	−13689484844281
c_{12}	42763026	c_{36}	−1366557808326266	c_{60}	−579353630072013952	c_{84}	1251284914822
c_{13}	328667290	c_{37}	3421562757891643	c_{61}	23794744423739277	c_{85}	1209782029811
c_{14}	−1118233743	c_{38}	1526346418792450	c_{62}	469193217069584125	c_{86}	−69724343416
c_{15}	−2820849678	c_{39}	−2306293751427560	c_{63}	−464000857099608630	c_{87}	−142218667068
c_{16}	331765972	c_{40}	−6166296900825113	c_{64}	400769826209161216	c_{88}	48748420925
c_{17}	13303424547	c_{41}	13898957594341763	c_{65}	−105296628146452307	c_{89}	−2732375766
c_{18}	59234060163	c_{42}	−237459118534556	c_{66}	−129004672693864301	c_{90}	−2144981360
c_{19}	−94290947997	c_{43}	−13067697628619274	c_{67}	159389350192729849	c_{91}	222730600
c_{20}	−69977433451	c_{44}	−28844428466906634	c_{68}	−118387243579828144	c_{92}	−1849232
c_{21}	−612318714204	c_{45}	−16251898406616638	c_{69}	22048592861334756	c_{93}	−120064
c_{22}	1497217193214	c_{46}	136817106241414125	c_{70}	−5850781138269055	c_{94}	8960
c_{23}	−394113138682	c_{47}	−51577488605498030	c_{71}	18706196120201234		
c_{24}	3064969116977	c_{48}	73721144085594691	c_{72}	9248281107964690		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 94, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $94 + 4$ terms removed returns order 94 as well, so $\deg q_{\min} = 94 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $94 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 94 that this sequence satisfies — and that family turns out to have exactly one member.

References

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